Random sample: Any set of reliazations $(X_1, X_2, ..., X_n)$ made on X under independent and identical conditions is called a **random sample**.

Statistic: Any statistical quantity calculated on the basis of the random sample is called a **statistic**. The sample mean, sample standard deviation, sample proportion *etc*., are called **statistics** (plural form of *statistic*).They will be denoted by Roman letters.

Let $(x_1, x_2, \ldots, x_n)$ be an observed value of $(X_1, X_2, ..., X_n)$. The collection of $(x_1, x_2, \ldots, x_n)$ is known as *sample space*, which will be denoted by '**S**'.

Note 1:

A set of n sample observations can be made on X, say, $x_1, x_2, \ldots, x_n$ for making inferences on the unknown parameters. It is to be noted that these n values may vary from sample to sample. Thus, these values can be considered as the realizations of the random variables $X_1, X_2, ..., X_n$, which are assumed to be independent and have the same distribution as that of X. These are also called independently and identically distributed (*iid)* random variables.

NOTE

The *statistic* itself is a random variable and has a probability distribution.

Note 2:

In Statistical Inference, the sample standard deviation is defined as $S = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2}$, where $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$. It may be noted that the divisor is $n - 1$ instead of n.

Note 3:

The statistic itself is a random variable, until the numerical values of $X_1, X_2, ..., X_n$ are observed, and hence it has a probability distribution.

Notations to denote various population parameters and their corresponding sample statistics are listed in Table 1.1. The notations will be used in the first four chapters of this book with the same meaning for the sake of uniformity.

Table 1.1 Notations for Parameters and Statistics

Statistical measure	Parameter	Statistic	Value of the Statistic for a given sample
Mean	μ	$\bar{X}$	$\bar{x}$
Standard deviation	σ	S	s
Proportion	P	p	p_0

1.2 SAMPLING DISTRIBUTION

The probability distribution of a statistic is called **sampling distribution** of the statistic. In other words, it is the probability distribution of possible values of the statistic, whose values are computed from possible random samples of same size.

The following example will help to understand this concept.

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